

Ontario Heart Monitoring Analytics



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Content Outline

Topics for discussion

- 01 Introduction to the Company
- 02 Data Set from 2010 to 2016
- 03 Data Creation Using : Vectors, Matrices and Data Frames
- 04 Analysis: DPLYR Library
- 05 Visualization: Graphical Representation



OHMA

Data Saves Lives.

Ontario Heart Monitoring Analytics (OHMA) was founded in 2015 by 5 classmates working on a project. Little did they know that 6 years down the lane, OHMA would play a vital part in helping the people of Ontario to lead a healthier lifestyle.

OHMA runs on the basis of gathering heart diseases related data by collecting information depending on gender, age, year, population.



Dataset (2010 – 2016)

	A	B	C	D	E	F	G	H
1	Age Group	Sex	Year	Rate (per 100,000)	Number	Population		
2	'20+'	Both sexes	2010	224	23,470	10,492,970		
3	'20-34'	Both sexes	2010	4	110	2,730,600		
4	'35-49'	Both sexes	2010	63	1,970	3,144,930		
5	'50-64'	Both sexes	2010	246	6,660	2,711,710		
6	'65-79'	Both sexes	2010	553	7,420	1,342,670		
7	'80+'	Both sexes	2010	1,300	7,320	563,080		
8	'20+'	Females	2010	168	9,110	5,408,000		
9	'20-34'	Females	2010	1	20	1,369,270		
10	'35-49'	Females	2010	28	450	1,581,130		
11	'50-64'	Females	2010	123	1,710	1,385,210		
12	'65-79'	Females	2010	400	2,880	720,770		
13	'80+'	Females	2010	1,149	4,040	351,620		
14	'20+'	Males	2010	282	14,360	5,084,970		
15	'20-34'	Males	2010	6	80	1,361,320		
16	'35-49'	Males	2010	97	1,520	1,563,800		
17	'50-64'	Males	2010	372	4,940	1,326,480		
18	'65-79'	Males	2010	730	4,540	621,900		
19	'80+'	Males	2010	1,546	3,270	211,460		
20	'20+'	Both sexes	2011	213	22,700	10,674,590		
21	'20-34'	Both sexes	2011	4	110	2,779,460		

heartdata
+

Ready

Creating Vectors and the Data Frame

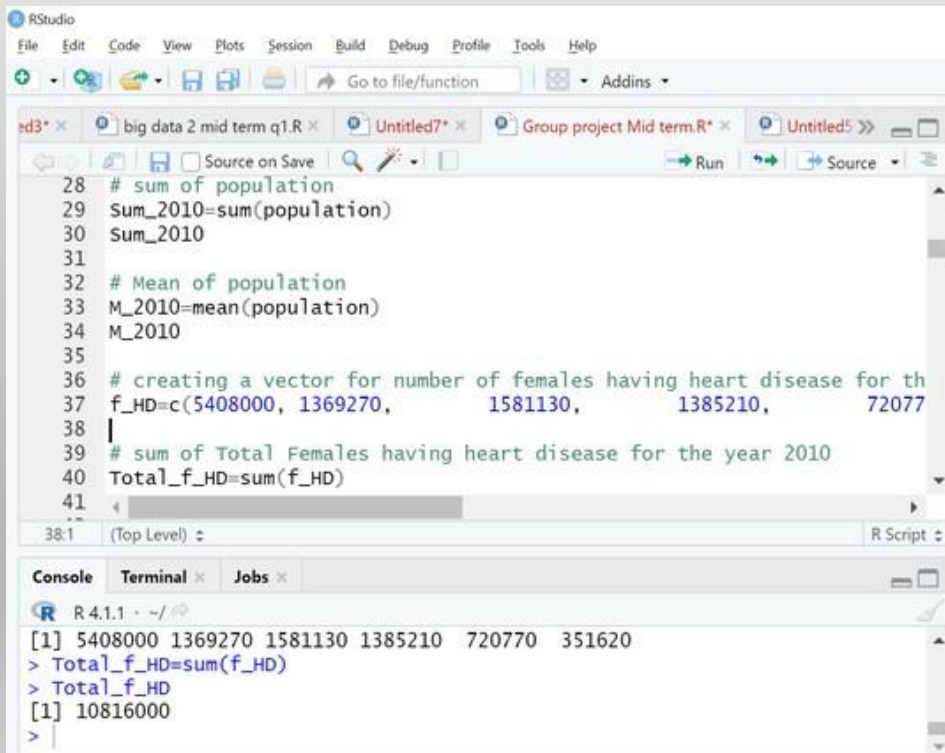
```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins
led3* x big data 2 mid term q1.R x Untitled7* x Group project Mid term.R* x Untitled7* x
Source on Save Run Source
7 #creating vectors for variables/columns agegroup,sex,Year,Xrateper100
8
9 agegroup<-c('20+', '20-34', '35-49', '50-64', '65-79', '80+')
10
11 sex<-c('Both Sexes','Male','Female')
12
13 Year =c('2010')
14
15 Xrateper100<-c(224, 4, 63, 246, 553, 1300, 168,
16
17 number<-c(23470, 110, 1970, 6660, 7420, 7320, 9110,
18
12:1 (Top Level) R Script
```

```
R 4.1.1 - ~/
> Xrateper100
[1] 224 4 63 246 553 1300 168 1 28 123 400 1149 282
[14] 6 97 372 730 1546
> number
[1] 23470 110 1970 6660 7420 7320 9110 20 450 1710 2880
[12] 4040 14360 80 1520 4940 4540 3270
> |
```

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins
led3* x big data 2 mid term q1.R x Untitled7* x Group project Mid term.R* x Untitled7* x
Source on Save Run Source
22
23 # creating a data frame by using above vectors
24 heartdata <-data.frame(agegroup,sex,Year,Xrateper100,number,population)
25
26 heartdata
27
28 # sum of population
29 Sum_2010=sum(population)
30 Sum_2010
31
32 # Mean of population
33
8:1 (Top Level) R Script
```

```
R 4.1.1 - ~/
> heartdata <-data.frame(agegroup,sex,Year,Xrateper100,number,population)
> heartdata
  agegroup      sex Year Xrateper100 number population
1    20+ Both Sexes 2010      224  23470  10492970
2    20-34      Male 2010         4    110   2730600
3    35-49    Female 2010        63   1970  3144930
4    50-64 Both Sexes 2010      246   6660  2711710
5    65-79      Male 2010      553   7420  1342670
```

Using Sum function on Vector for Females and Males

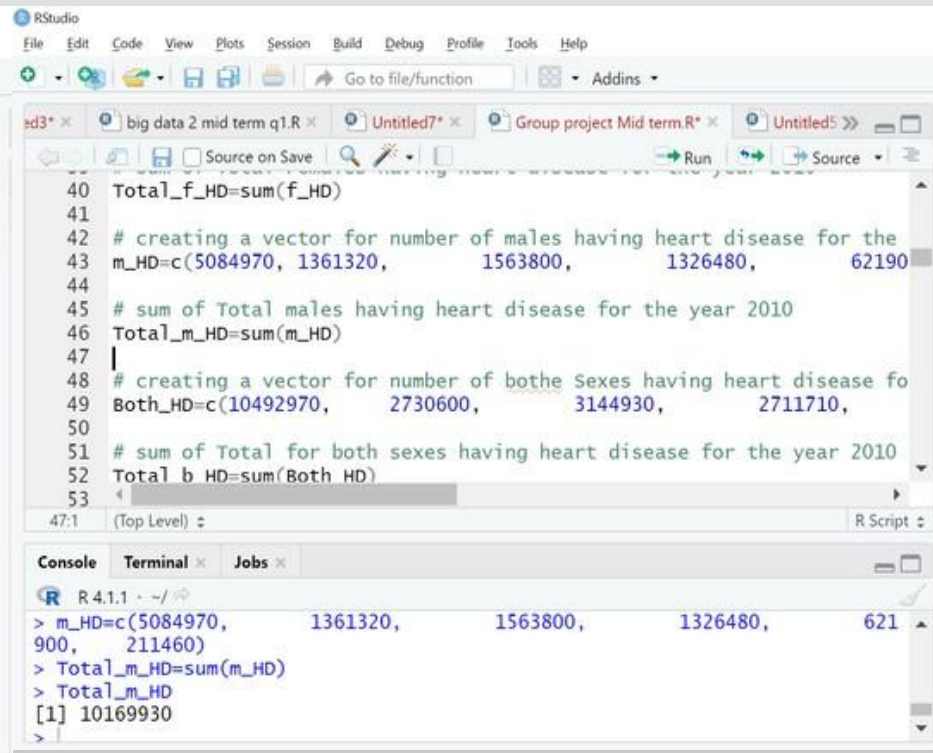


The RStudio interface shows a script with the following code:

```
28 # sum of population
29 Sum_2010=sum(population)
30 Sum_2010
31
32 # Mean of population
33 M_2010=mean(population)
34 M_2010
35
36 # creating a vector for number of females having heart disease for th
37 f_HD=c(5408000, 1369270, 1581130, 1385210, 72077
38 |
39 # sum of Total Females having heart disease for the year 2010
40 Total_f_HD=sum(f_HD)
41
```

The console output shows the results of the calculations:

```
R 4.1.1 ~ ./
[1] 5408000 1369270 1581130 1385210 72077 351620
> Total_f_HD=sum(f_HD)
> Total_f_HD
[1] 10816000
>
```



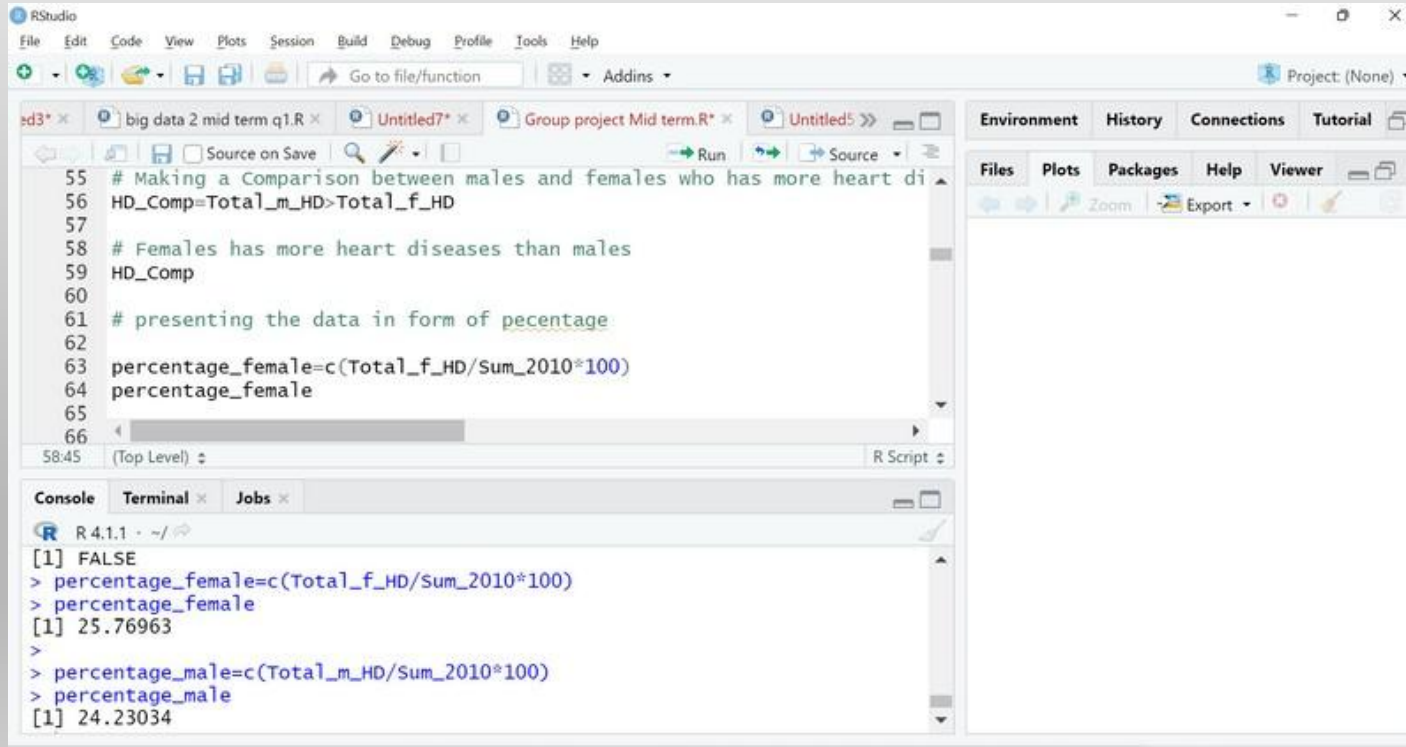
The RStudio interface shows a script with the following code:

```
40 Total_f_HD=sum(f_HD)
41
42 # creating a vector for number of males having heart disease for the
43 m_HD=c(5084970, 1361320, 1563800, 1326480, 62190
44
45 # sum of Total males having heart disease for the year 2010
46 Total_m_HD=sum(m_HD)
47 |
48 # creating a vector for number of both sexes having heart disease fo
49 Both_HD=c(10492970, 2730600, 3144930, 2711710,
50
51 # sum of Total for both sexes having heart disease for the year 2010
52 Total b HD=sum(Both HD)
53
```

The console output shows the results of the calculations:

```
R 4.1.1 ~ ./
> m_HD=c(5084970, 1361320, 1563800, 1326480, 621
900, 211460)
> Total_m_HD=sum(m_HD)
> Total_m_HD
[1] 10169930
>
```

Making Comparison between males and females



The screenshot shows the RStudio interface with the following components:

- Source Editor:** Contains R code for comparing heart diseases between males and females.
- Console:** Shows the execution output of the code.
- Environment/Plots/Packages/Help/Viewer:** The right-hand pane, currently empty.

Source Editor Code:

```
55 # Making a Comparison between males and females who has more heart di
56 HD_Comp=Total_m_HD>Total_f_HD
57
58 # Females has more heart diseases than males
59 HD_Comp
60
61 # presenting the data in form of percentage
62
63 percentage_female=c(Total_f_HD/Sum_2010*100)
64 percentage_female
65
66
```

Console Output:

```
R 4.1.1 ~ /
[1] FALSE
> percentage_female=c(Total_f_HD/Sum_2010*100)
> percentage_female
[1] 25.76963
>
> percentage_male=c(Total_m_HD/Sum_2010*100)
> percentage_male
[1] 24.23034
```


Creating a new single Matrix by Combining Column Matrices

The screenshot shows the RStudio IDE interface. The top menu bar includes File, Edit, Code, View, Plots, Session, Build, Debug, Profile, Tools, and Help. The toolbar contains icons for file operations and a search bar labeled 'Go to file/function'. The main editor window displays R code for creating matrices. The console window at the bottom shows the execution of the code, including the creation of the 'Xrateper100' matrix.

R Code in Editor:

```

70 # #creating Matrices for variables/Columns agegroup,sex,Year,Xratepe
71
72 agegroup<-matrix(c('20+', '20-34', '35-49', '50-64', '65-79', '80+'), nrow
73
74 sex<-matrix(c('Both Sexes', 'Male', 'Female'), nrow = 18, byrow = TRUE)
75
76 Year =matrix(c('2016'), nrow = 18, byrow = TRUE)
77
78 Xrateper100<-matrix(c(198,      4,      53,      219,      451,      1016,
79 886,      260,      7,      85,      335,      619,      1219), nrow = 18, byr
80

```

Console Output:

```

R 4.1.1 ~ /
> agegroup<-matrix(c('20+', '20-34', '35-49', '50-64', '65-79', '80+'), nrow = 1
8, byrow = TRUE)
> sex<-matrix(c('Both Sexes', 'Male', 'Female'), nrow = 18, byrow = TRUE)
> Year =matrix(c('2016'), nrow = 18, byrow = TRUE)
> Xrateper100<-matrix(c(198,      4,      53,      219,      451,      1016,
1,      2,      22,      108,      304,
+ 886,      260,      7,      85,      335,      619,      1219), nrow = 18, byrow = T
RUE)
> Xrateper100<-matrix(c(198,      4,      53,      219,      451,      1016,
120,      1600,      6770,      7780,      6470,      830

```

The screenshot shows the RStudio interface. The script editor contains the following R code:

```

86
87 # combining the Matrices into a single matrix
88 B_Matrix=cbind(agegroup,sex,Year,Xrateper100,number,population_2016)
89
90 # Naming the column of the matrix
91 col <-c ("agegroup","sex","Year","Xrateper100","number","population_2
92
93 colnames(B_Matrix) <-col
94
95 print(B_Matrix)
96
97

```

The console shows the output of the code:

```

R 4.1.1 - ~/
> col <-c ("agegroup","sex","Year","Xrateper100","number","population_2016")
> colnames(B_Matrix) <-col
> print(B_Matrix)
  agegroup sex      Year Xrateper100 number population_2016
[1,] "20+"   "Both Sexes" "2016" "198"      "22730" "11454740"
[2,] "20-34"  "Male"      "2016" "4"         "120"   "2971870"
[3,] "35-49"  "Female"    "2016" "53"        "1600"  "3030230"

```


DPLYR

By Using Select Function output

```
6 mydata2
7 # Showing data without Year and Population columns
8 mydataminus = select(mydata, -Year, -Population)
9 mydataminus
10 # Selecting variables contain 'x' in their columns names
11 mydata_X = select(mydata, starts_with("X"))
12 mydata_X
```

12:40 (Top Level) ↕

Console Terminal x Jobs x

R 4.1.1 · C:/Users/ontpn/OneDrive/Desktop/Humber College/Smester 3/big data 2/ ↗

126 2016 249310

```
> mydataminus = select(mydata, -Year, -Population)
```

```
> mydataminus
```

	Age.Group	Sex	X.Rate..per.100.000..	Number
1	'20+'	Both sexes	224	23470
2	'20-34'	Both sexes	4	110
3	'35-49'	Both sexes	63	1970
4	'50-64'	Both sexes	246	6660
5	'65-79'	Both sexes	553	7420
6	'80+'	Both sexes	1300	7320
7	'20+'	Females	168	9110
8	'20-34'	Females	1	20
9	'35-49'	Females	28	450
10	'50-64'	Females	123	1710
11	'65-79'	Females	400	2880
12	'80+'	Females	1149	4040
13	'20+'	Males	282	14360
14	'20-34'	Males	6	80
15	'35-49'	Males	97	1520

```
10 # Selecting variables contain 'B' and star with 'x' in their columns name
11 mydata_X = select(mydata, starts_with("X"))
12 mydata_X
13 mydata_B = select(mydata, contains("B"))
14 mydata_B
15
```

12:9 (Top Level) ↕

Console Terminal x Jobs x

R 4.1.1 · C:/Users/ontpn/OneDrive/Desktop/Humber College/Smester 3/big data 2/ ↗

X.Rate..per.100.000..

1	224
2	4
3	63
4	246
5	553
6	1300
7	168
8	1
9	28
10	123
11	400
12	1149
13	282
14	6
15	97
16	372
17	730
18	1546
19	213
20	4
21	59
22	223
23	519
24	1243

Using Arrange and Distinct function

Group project Mid term.R* x Untitled13* x pie chart.R x heart_data_ontario11 x

Source on Save

```
13 mydata_B = select(mydata, contains('B'))
14 mydata_B
15
16 # using Arrange function
17 arrange_mydata=arrange(mydata, desc(Year))
18
```

17:1 (Top Level) ↓

Console Terminal x Jobs x

R 4.1.1 · C:/Users/ontpn/OneDrive/Desktop/Humber College/Semster 3/big data 2/ ↗

```
> arrange_mydata
  Age.Group      Sex Year X.Rate..per.100.000.. Number Population
1 '20+' Both sexes 2016      198    22730    11454740
2 '20-34' Both sexes 2016         4      120     2971870
3 '35-49' Both sexes 2016        53     1600    3030230
4 '50-64' Both sexes 2016       219     6770    3089960
5 '65-79' Both sexes 2016       451     7780    1726160
6 '80+' Both sexes 2016      1016     6470     636520
7 '20+' Females 2016       141     8300    5897690
8 '20-34' Females 2016         2        30    1474170
9 '35-49' Females 2016        22       340    1540860
10 '50-64' Females 2016       108     1690    1571070
11 '65-79' Females 2016       304     2810     924360
12 '80+' Females 2016       886     3430     387210
13 '20+' Males 2016       260    14430    5557050
14 '20-34' Males 2016         7       100    1497710
15 '35-49' Males 2016        85     1260    1489370
16 '50-64' Males 2016       335     5090    1518890
17 '65-79' Males 2016       619     4960     801790
```

19
20 #Below we are using two variables Sex and Year to determine uniqueness
21 x5 = distinct(mydata, Sex, Year, keep_all= TRUE)
22
23
24
20:7 (Top Level) ↓

Console Terminal x Jobs x

R 4.1.1 · C:/Users/ontpn/OneDrive/Desktop/Humber College/Semster 3/big data 2/ ↗

```
4 2013 TRUE
5 2014 TRUE
6 2015 TRUE
7 2016 TRUE
> x5 = distinct(mydata, Sex, Year, keep_all= TRUE)
> x5
      Sex Year keep_all
1 Both sexes 2010    TRUE
2 Females 2010    TRUE
3 Males 2010    TRUE
4 Both sexes 2011    TRUE
5 Females 2011    TRUE
6 Males 2011    TRUE
7 Both sexes 2012    TRUE
8 Females 2012    TRUE
9 Males 2012    TRUE
10 Both sexes 2013    TRUE
11 Females 2013    TRUE
12 Males 2013    TRUE
13 Both sexes 2014    TRUE
14 Females 2014    TRUE
```

Filter, Pipe and Grepl function

```
35 mydata8 = filter(mydata, Sex == Males )
36 # using Pipe and filter getting data of specific column
37 filter_spec = filter(mydata, Sex %in% c("Males") & Year %in% c("2010") & Age.Group %in% c("'80+"
38 filter_spec
39 # using Filter and grepl function to show value in population column having figure 23
40 mydata10 = filter(mydata,grepl("23",Population))
41 |
42 |
```

41:1 (Top Level) ↕

R Script

Console

Terminal ×

Jobs ×

R 4.1.1 · C:/Users/ontpn/OneDrive/Desktop/Humber College/Smester 3/big data 2/ ↗

```
31      20+ Males 2015      237 14000 3403390
32      '20-34' Males 2015      5      80 1471510
33      '35-49' Males 2015      88 1300 1484250
34      '50-64' Males 2015     328 4910 1497500
35      '65-79' Males 2015     624 4820 772470
36      '80+' Males 2015    1239 2970 239670
37      '20+' Males 2016     260 14430 5557050
38      '20-34' Males 2016      7 100 1497710
39      '35-49' Males 2016     85 1260 1489370
40      '50-64' Males 2016     335 5090 1518890
41      '65-79' Males 2016     619 4960 801790
42      '80+' Males 2016    1219 3040 249310
```

```
> mydata8 = filter(mydata, Sex %in% c("Males") & Year %in% c("2010") & Age.Group %in% c("'80+"
> filter_spec = filter(mydata, Sex %in% c("Males") & Year %in% c("2010") & Age.Group %in% c("'80
+"))
> filter_spec
```

```
Age.Group Sex Year X.Rate..per.100.000.. Number Population
1 '80+' Males 2010 1546 3270 211460
```

```
> mydata10 = filter(mydata,grepl("23",Population))
```

```
> mydata10
```

```
Age.Group Sex Year X.Rate..per.100.000.. Number Population
1 '20-34' Females 2012 1 20 1411230
2 '80+' Males 2013 1333 3160 237120
3 '80+' Males 2014 1350 3180 235490
```

Adding New Variable By Using Mutate Function

```
50  
57  
58  
59 # percentage of affected people from heart disease by adding new variable  
60 percent_Ag=mydata %>% select(Number,Population) %>% mutate(mydata,percent_Ag=(Number/Population  
61 percent_Ag)
```

61:11 (Top Level) ↕

R Script ↕

Console

Terminal x

Jobs x

R 4.1.1 · C:/Users/ontpn/OneDrive/Desktop/Humber College/Semester 3/big data 2/ ↗

```
> percent_Ag=mydata %>% select(Number,Population) %>% mutate(mydata,percent_Ag=(Number/Population*100))
```

```
> percent_Ag
```

	Number	Population	Age.Group	Sex	Year	X.Rate..per.100.000..	percent_Ag
1	23470	10492970	'20+'	Both sexes	2010	224	0.223673564
2	110	2730600	'20-34'	Both sexes	2010	4	0.004028419
3	1970	3144930	'35-49'	Both sexes	2010	63	0.062640504
4	6660	2711710	'50-64'	Both sexes	2010	246	0.245601484
5	7420	1342670	'65-79'	Both sexes	2010	553	0.552630207
6	7320	563080	'80+'	Both sexes	2010	1300	1.299992896
7	9110	5408000	'20+'	Females	2010	168	0.168454142
8	20	1369270	'20-34'	Females	2010	1	0.001460632
9	450	1581130	'35-49'	Females	2010	28	0.028460658
10	1710	1385210	'50-64'	Females	2010	123	0.123446986

Using Rank Function In Descending Order

```
170 mydata$Number_rank=rank(mydata$Number)
171 mydata$Number_rank
172
173 # Using Rank Function with Descending order on Year Column
174 mydata$Year_rank=rank(desc(mydata$Year))
175 mydata$Year_rank
176
177
```

```
173 # Using Rank Function with Descending order on Year Column
174 mydata$Year_rank=rank(desc(mydata$Year))
175 mydata$Year_rank
176
```

176
177
177:1 (Top Level) R Script

177:1	(Top Level) ⌵	R Scrip
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177:1	(Top Level) ⌵	R Scrip
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Console Terminal  Jobs 

Console Terminal  Jobs 

Console Terminal  Jobs 

R 4.1.1 · C:/Users/ontpn/OneDrive/Desktop/Humber College/Smester 3/big data 2/ ↗

123	1260	1489370	'35-49'	Males	2016	85	0.084599529
124	5090	1518890	'50-64'	Males	2016	335	0.335113142
125	4960	801790	'65-79'	Males	2016	619	0.618615847
126	3040	249310	'80+'	Males	2016	1219	1.219365449

```
> mydata$Number_rank=rank(mydata$Number)
```

```
> mydata$Number_rank
```

[1]	126.0	16.5	49.0	92.0	100.0	99.0	112.0	2.5	28.0	44.0	53.0	70.0	117.5	9.0	35.0
[16]	82.0	72.0	63.0	121.0	16.5	48.0	85.0	98.0	97.0	109.5	6.0	26.5	36.0	50.0	69.0
[31]	113.0	9.0	34.0	74.0	71.0	61.0	123.0	16.5	47.0	86.0	101.0	96.0	111.0	2.5	26.5
[46]	38.5	56.0	68.0	114.0	11.0	33.0	75.0	73.0	62.0	125.0	20.0	46.0	90.0	103.0	95.0
[61]	109.5	2.5	25.0	42.0	55.0	67.0	116.0	13.0	32.0	79.0	76.0	59.0	124.0	20.0	42.0
[76]	89.0	105.0	94.0	108.0	2.5	23.5	40.0	54.0	66.0	117.5	13.0	31.0	78.0	80.5	60.0
[91]	120.0	16.5	38.5	91.0	102.0	87.0	107.0	6.0	23.5	45.0	52.0	65.0	115.0	9.0	30.0
[106]	80.5	77.0	57.0	122.0	20.0	37.0	93.0	104.0	88.0	106.0	6.0	22.0	42.0	51.0	64.0
[121]	119.0	13.0	29.0	84.0	83.0	58.0									

```
> mydata$Year_rank=rank(desc(mydata$Year))
```

```
> mydata$Year_rank
```

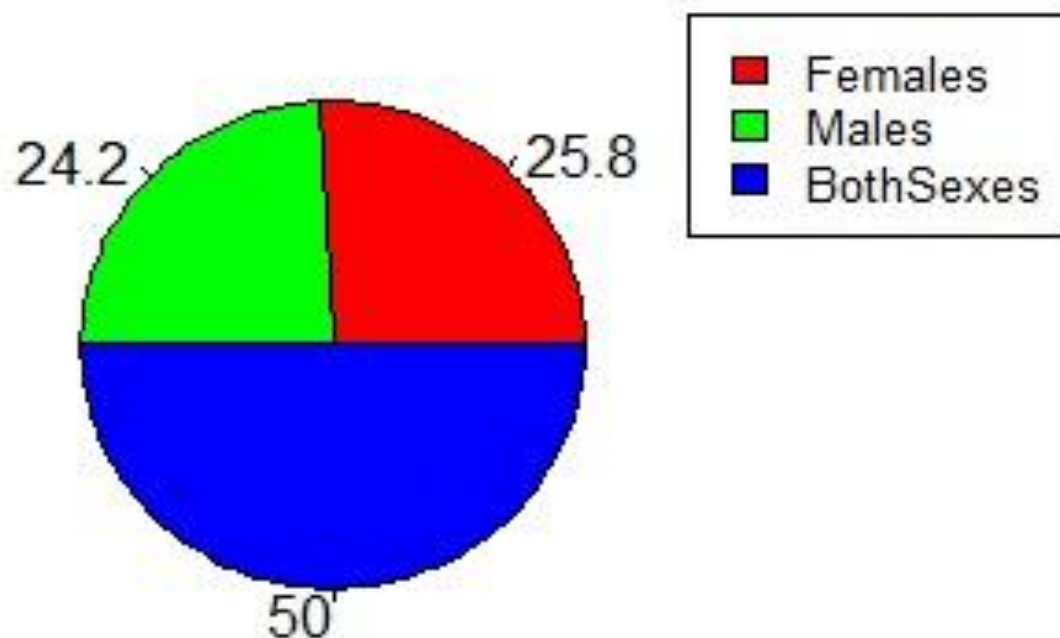
[illegible]

GRAPHS

Pie Chart : Heart Diseases percentage by Sex

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins
dplyr tutorial.R x ass mid ali.R x Group project Mid term.R x graphs edward (1).R x
Source on Save Run Source
179
180 pop <- c(Total_f_HD,Total_m_HD,Total_b_HD)
181 labels <- c("Females", "Males","BothSexes")
182
183 piepercent<- round(100*pop/sum(pop), 1)
184 # plot the chart.
185 pie(pop, labels = piepercent, main = "Percentage of HeartDisease By sex",col = rainbow(length(pop)),
186 legend("topright", c("Females", "Males","BothSexes"), cex = 0.8,fill = rainbow(length(pop)))
187
188
186:1 (Top Level) R Script
Console Terminal x Jobs x
R 4.1.1 ~
> # Plot the chart.
> pie(pop, labels = piepercent, main = "population pie chart",col = rainbow(length(pop)))
> legend("topright", c("Females", "Males","BothSexes"), cex = 0.8,
+ fill = rainbow(length(pop)))
> # Save the file.
> dev.off()
RStudioGD
2
>
> mydata = read.csv("C:\\Users\\91873\\Desktop\\Humber\\SEMESTER 3\\Big data 2\\heart _data _ontari
o.csv")
> pop <- c(Total_f_HD,Total_m_HD,Total_b_HD)
> labels <- c("Females", "Males","BothSexes")
> piepercent<- round(100*pop/sum(pop), 1)
> # Plot the chart.
> pie(pop, labels = piepercent, main = "Percentage of HeartDisease By sex",col = rainbow(length(po
p)))
>
```

Perctentage of HeartDisease By sex



Rate of Heart Diseases in Ontario per Year differentiated by Gender

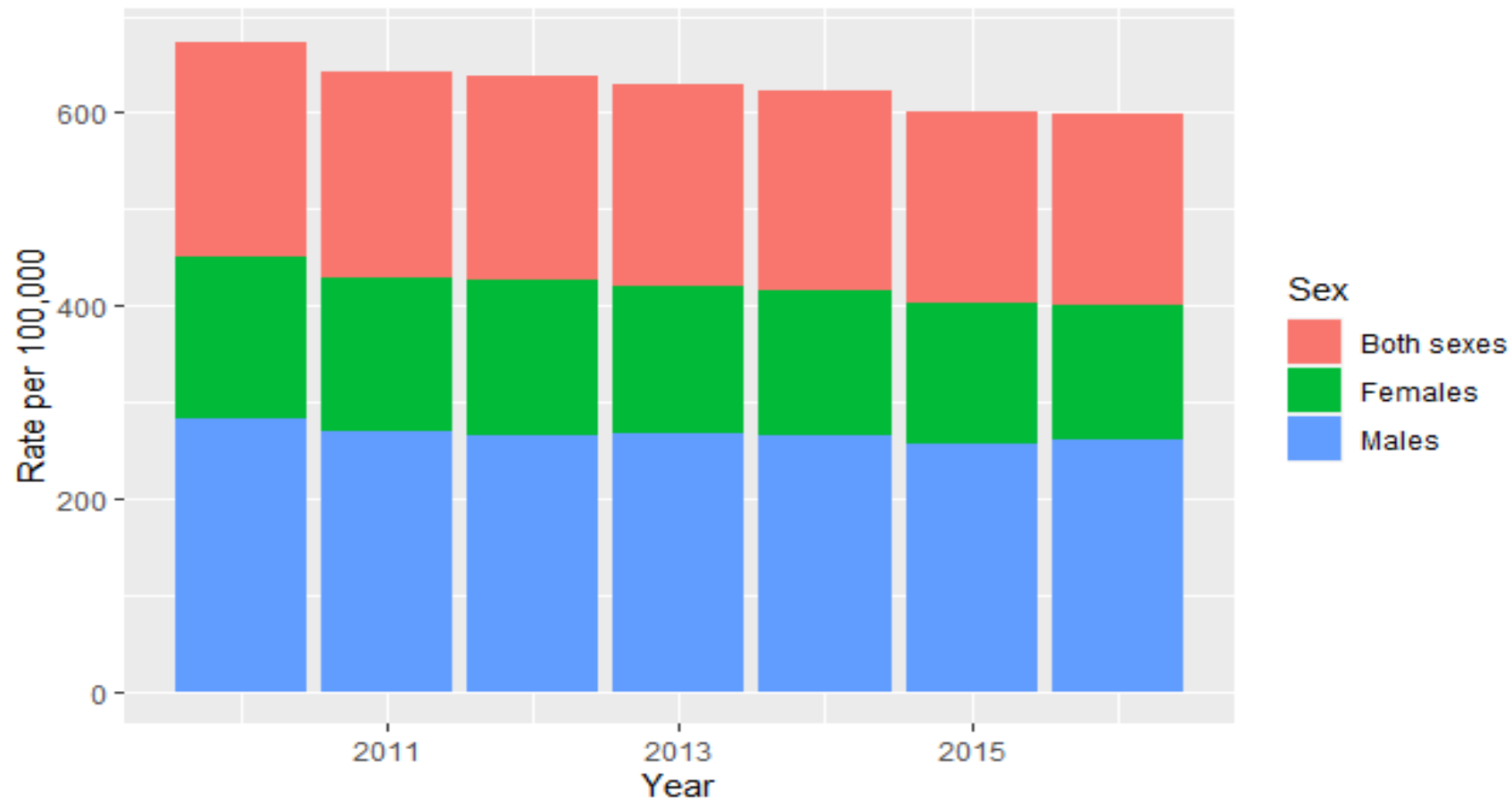
```
28 # Ontario Heart Rate data per Year
29
30 year <- subset(heartdata, Age.Group == "20+", select = c("Year","Sex","Rate"))
31
32 ggplot(year, aes(x=Year, y=Rate, fill=Sex)) +
33   geom_histogram(stat='identity')+
34   labs(title = "Heart disease in Ontario per year",x="Age Group",y="Rate per 100,000")
35
36
37
38
```

37:1 (Top Level) R Script

Console Terminal Jobs

```
R 4.1.1 ~ /
+   labs(title = "Heart disease in Ontario per year",x="Age Group",y="Rate per 100,000")
Warning message:
Ignoring unknown parameters: binwidth, bins, pad
> # Ontario Heart Rate data per Year
>
> year <- subset(heartdata, Age.Group == "20+", select = c("Year","Sex", "Rate"))
>
> ggplot(year, aes(x=Year, y=Rate, fill=Sex)) +
+   geom_histogram(stat='identity')+
+   labs(title = "Heart disease in Ontario per year",x="Age Group",y="Rate per 100,000")
Warning message:
Ignoring unknown parameters: binwidth, bins, pad
> total <- subset(heartdata, Age.Group == "20+", select = c("Age.Group","Year","Rate"))
>
> ggplot(total, aes(x=Year, y=Rate)) +
+   geom_bar(stat='identity') +
+   labs(title = "Yearly total number of heart disease in Ontario",x="Year",y="Rate per 100,000")
> total <- subset(heartdata, Age.Group == "20+", select = c("Age.Group","Year","Rate"))
>
> ggplot(total, aes(x=Year, y=Rate)) +
+   geom_bar(stat='identity') +
+   labs(title = "Yearly total number of heart disease in Ontario",x="Year",y="Rate per 100,000")
> year <- subset(heartdata, Age.Group == "20+", select = c("Year","Sex","Rate"))
>
> ggplot(year, aes(x=Year, y=Rate, fill=Sex)) +
+   geom_histogram(stat='identity')+
+   labs(title = "Heart disease in Ontario per year",x="Age Group",y="Rate per 100,000")
Warning message:
Ignoring unknown parameters: binwidth, bins, pad
>
```

Heart disease in Ontario per year



Total Number of Heart Disease in Ontario (Yearly)

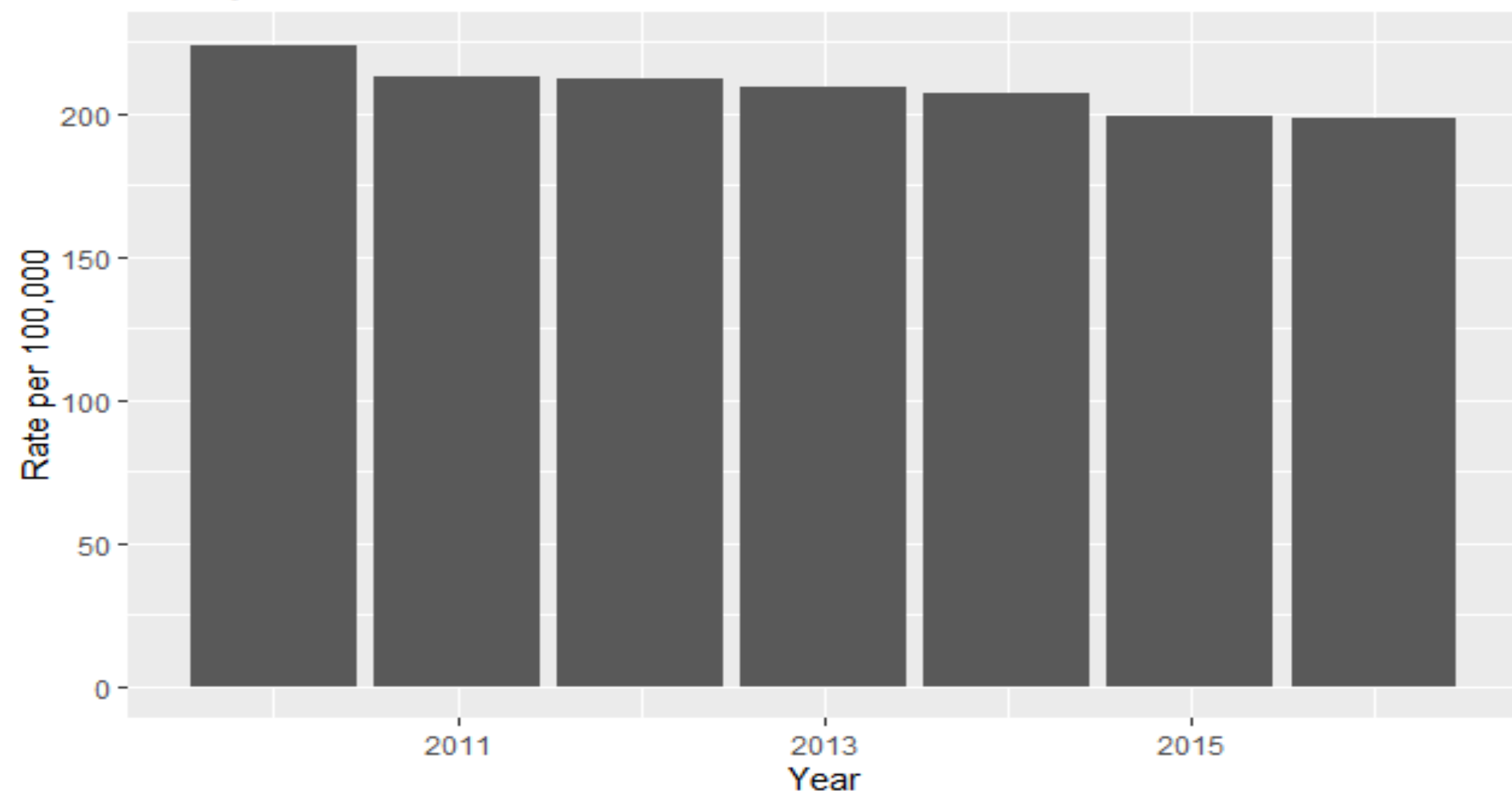
```
37
38 total <- subset(heartdata, Sex=="Both sexes", select = c("Age.Group","Sex","Year","Rate"))
39
40 totalYear <- filter(total, Age.Group == "20+" )
41
42 ggplot(totalYear, aes(x=Year, y=Rate)) +
43   geom_bar(stat='identity') +
44   labs(title = "Yearly total number of heart disease in Ontario",x="Year",y="Rate per 100,000")
45
```

40:26 (Top Level) R Script

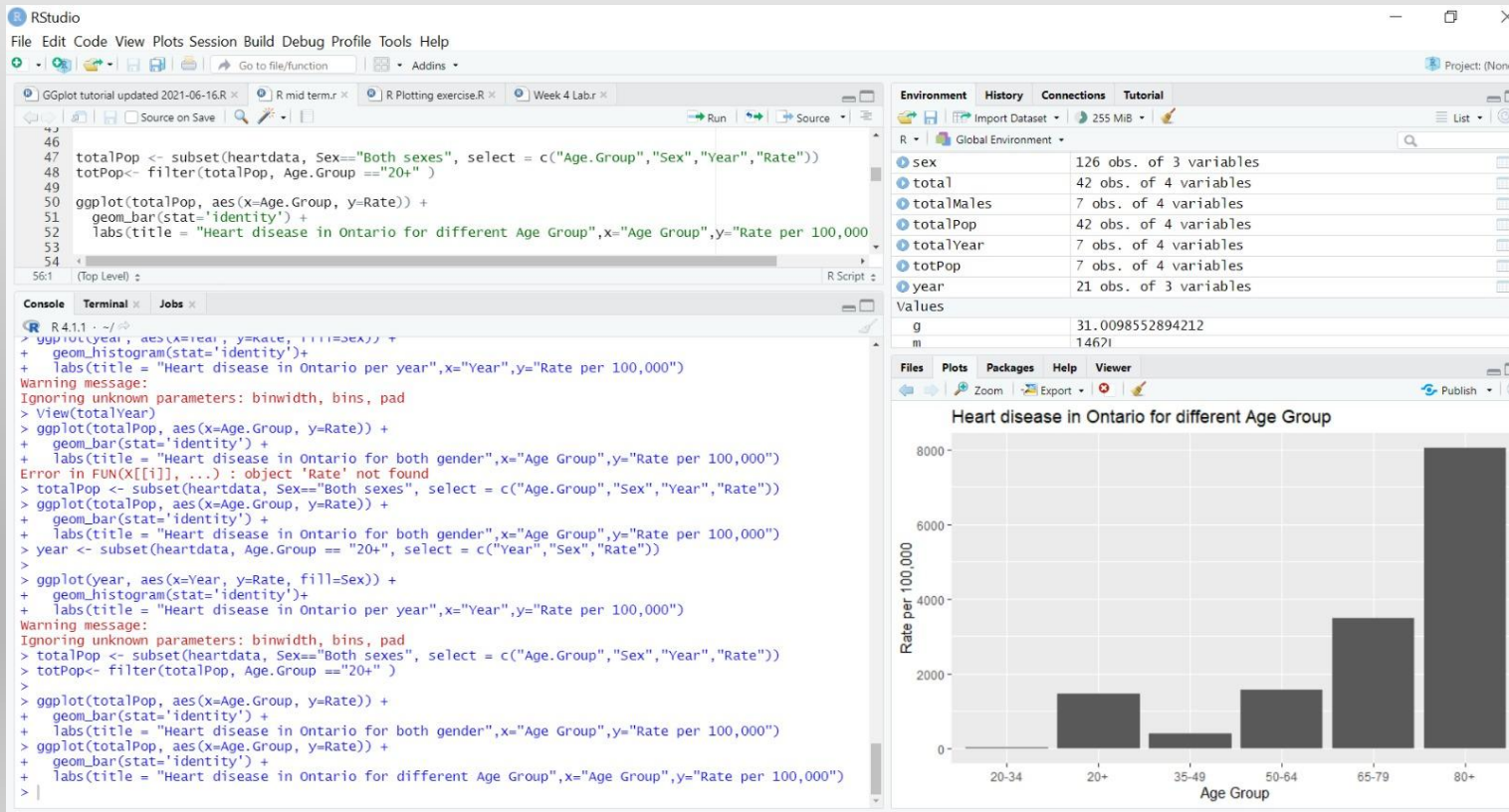
Console Terminal Jobs

```
R 4.1.1 ~ /
> ggplot(total, aes(x=Year, y=Rate)) +
+   labs(title = "Yearly total number of heart disease in Ontario",x="Year",y="Rate per 100,000")
> view(total)
> view(total)
> total <- subset(heartdata, Sex="Both sexes", Age.Group=="20+", select = c("Age.Group","Sex","Year","Rate"))
> ggplot(total, aes(x=Year, y=Rate)) +
+   geom_bar(stat='identity') +
+   labs(title = "Yearly total number of heart disease in Ontario",x="Year",y="Rate per 100,000")
> view(total)
> view(total)
> total <- subset(heartdata, Sex=="Both sexes", Age.Group=="20+", select = c("Age.Group","Sex","Year","Rate"))
Error in `[.data.frame'](x, r, vars, drop = drop) :
  object 'Age.Group' not found
> ggplot(bothSex, aes(x=Age.Group, y=Rate)) +
+   geom_bar(stat='identity') +
+   labs(title = "Heart disease in Ontario for both gender",x="Age Group",y="Rate per 100,000")
> total <- subset(heartdata, Sex=="Both sexes", select = c("Age.Group","Sex","Year","Rate"))
>
> totalYear = filter(total, Age.Group == "20+" )
>
> ggplot(total, aes(x=Year, y=Rate)) +
+   geom_bar(stat='identity') +
+   labs(title = "Yearly total number of heart disease in Ontario",x="Year",y="Rate per 100,000")
> totalYear <- filter(total, Age.Group == "20+" )
> view(totalYear)
> view(totalYear)
> ggplot(totalYear, aes(x=Year, y=Rate)) +
+   geom_bar(stat='identity') +
+   labs(title = "Yearly total number of heart disease in Ontario",x="Year",y="Rate per 100,000")
>
```

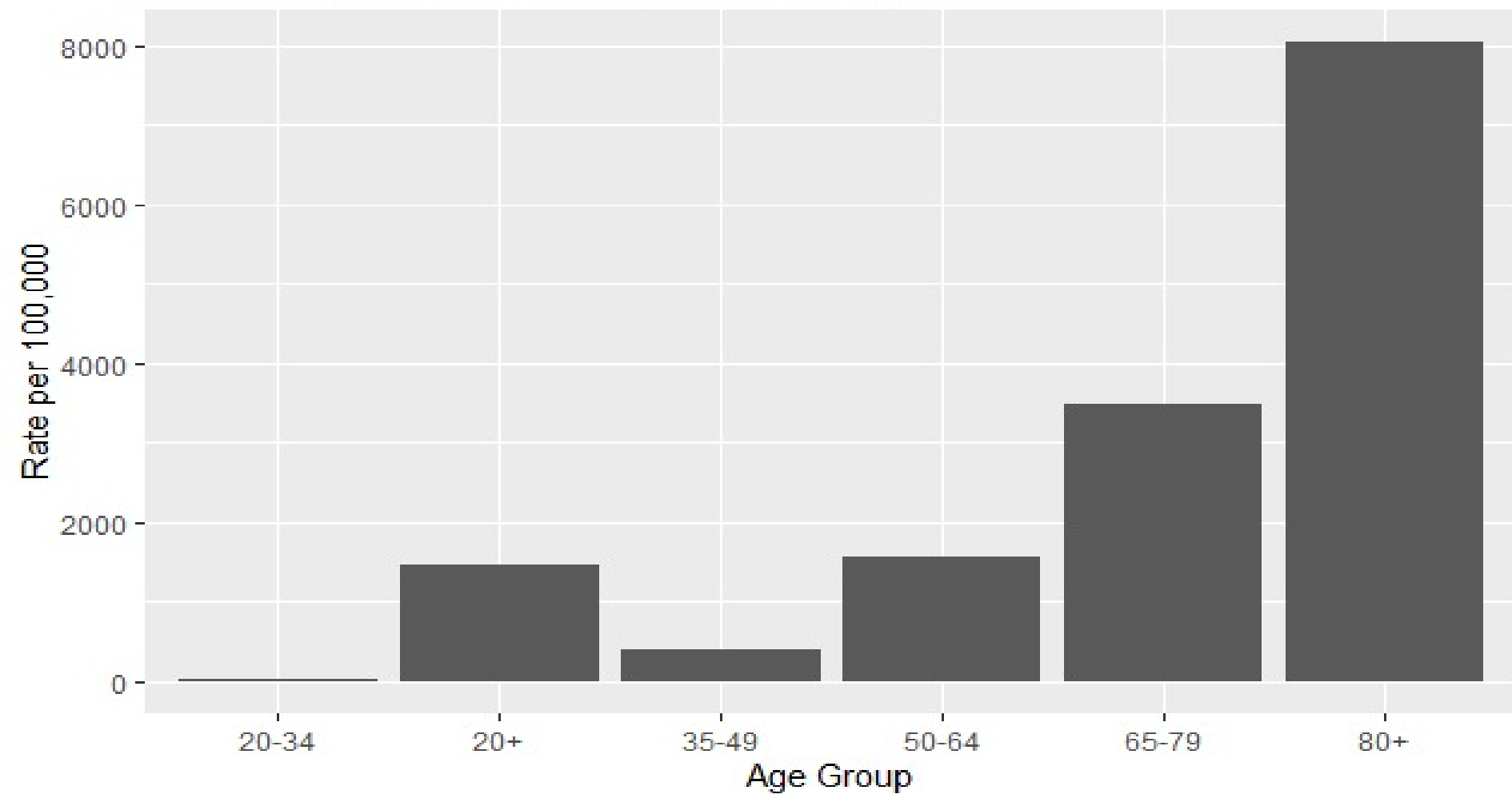
Yearly total number of heart disease in Ontario



Heart Diseases in Ontario by Age Group.



Heart disease in Ontario for different Age Group



Conclusion

- From the Histogram chart, we see that the % of heart disease is slightly higher in case of male than that of female. The rate of heart disease decreased from 2011 to 2015 gradually.
- It is interesting to find that middle age group has lower rate of heart disease as compared to aged groups (80+).
- It is also noteworthy over younger ages groups (20+) have higher tendency of heart disease possibly due to food habits.
- Overall, OHMA is trying to spread awareness in people about heart disease and in our analysis. People are now getting more conscious about their diet and exercise routine to live healthy life

Thanks for
attending.

Questions ?

